



ASTITVA JAISWAL

STUDENT OF COMPUTER SCIENCE AND ENGINEERING

Key skills: Pytorch, Python, Computer Vision, Transformers, Git, MySQL, MongoDB, Linux, Pandas, Prompt Engineering, Nvidia NeMo, JINA, nginx, Power BI, AWS, Sagemaker, Lightning

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https://ar5entum.vercel.app
Pune, Maharashtra, India



A highly skilled Research Engineer with hands-on experience in training and fine-tuning advanced language models and ASR and NLP systems, using cutting-edge technologies such as Huggingface Transformers, DeepSpeed, and Nvidia NeMo. Adept at deploying high-speed inference pipelines with a solid foundation in Machine Learning and Deep Learning techniques, complemented by strong backend development skills.

Professional Experience

Research Engineer

ASSISTO TECHNOLOGIES LTD, Noida, India

Pune
Jun 2024 - Present

- Trained **Speech to Text Wav2Vec2 models (94.4M and 300M parameters)** for low and high resource indic and foreign languages using **deepspeed GPU parallelization** using **torchrun** on **8 H100 GPUs** for Audio Datasets of **> 2000 hours** achieving **WER (Word Error Rate) of < 0.15 (85% accuracy)**
- Finetuned **seq2seq (Sequence to Sequence) language models** for **transliteration, spell correction and code mixing tasks** achieving **BLEU (Bilingual Evaluation Understudy) scores of 46.6** using **huggingface seq2seq MarianMT** on dataset of **60 million rows**.
- Trained language models **MarianMT (77.1M param)**, **BART (49.4M params)**, **MBART (611M params)** for tasks **text summarization, translation and spell correction**.
- Engineered and deployed models inference pipelines with **VAD (Voice Activation Detection)** and **GPU batch inference algorithms** for high speed inference (**30,000 minutes of audio per minute**)
- Wrote context based **text augmentation** algorithms for **ASR (Automatic Speech Recognition) error correction**.
- Dataset creation using **large language models (LLMs) OpenAI and LLaMa3** for **Intent recognition** and **NER (Named Entity Recognition)** for domain specific applications using **Prompt Engineering**.
- Actively contributing in R&D of **Generative AI (GenAI)** products with **large scale information retrieval** and **RAG (Retrieval Augmented Generation)** on **ElasticSearch** and in context AI agent integrations using **Langchain** and **LLaMa2 LLM**.

MLE Intern

Jan 2024 - Jun 2024

- Fine tuned **Encoder Decoder models** from huggingface transformers like **CharBERT** and **RoBERTa** for **Masked Language Modeling (MLM)** as a Seq2Seq for IPA and text phoneme correction. Fine tuned and deployed **text summarisation and QnA models using Longformers**.
 - Worked on **Flask RestAPIs and Jina** and assisted with **full scale python backend development using dockers, APIs, Databases etc**.
- Skills used:** Machine Learning · Deep Learning · Nvidia NeMo · Fine Tuning · Huggingface · Transformers · NLP · Wav2Vec · ASR · TTS LangChain · LLaMa2 · Elasticsearch · RAG · Prompt Engineering · nginx · Flask · Jina · RestAPI · websockets, matplotlib

Projects

Hindi to Hinglish (Devnagiri-Roman) Code Switch Translation

Personal

Pune
Aug 2024

- Developed a specialized machine learning model for **Hindi-English code-switching translation**. Successfully translated Devanagari script Hindi text into a mixed format, preserving Hindi words in Devanagari and converting English words to Roman script.
- Fine-tuned **MarianMT** based **Helsinki-NLP/opus-mt-mul-en** model on self curated and verified dataset of **100,000 row dataset (ar5entum/hindi-english-code-mixed)** achieving results on evaluation set with **Loss of 0.1450**, **BLEU score of 77.8649** and **Generation Length of 74.8945 in 20 epochs**. Open Sourced model on huggingface hub. (**ar5entum/marianMT_hin_eng_cs**)

Semantic Segmentation of Botanical Components with Computer Vision

Personal

Pune
Sept 2023

- Fine tuned models **Detectron2, YOLOv8, RCNNs** to **semantically segment** plant images into leaves, fruits, and flowers, addressing a challenge unmet by traditional methods. Dataset Creation: Created a manual dataset of **150 labeled plant images (20 validation)**.

Certifications

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

DeepLearning.AI

Online
Feb 2024

coursera.org/account/accomplishments/verify/5UGFDEZGFZPX

Neural Networks and Deep Learning

DeepLearning.AI

Online
Feb 2024

coursera.org/account/accomplishments/verify/SUMHRMN24URF

Publications

A Critical Analysis of Approaches to Glaucoma Detection

International Journal of Advanced Research in Science, Communication and Technology
DOI: 10.48175/IJARSCT-13871

ISSN: 2581-9429
Nov 2023

Education

MIT ADT University

BTech Computer Science and Engineering

CGPA: 7.65

2020 - 2024

City International School, Wanowrie

12th CBSE, AISSCE

Percentage: 80%

2017 - 2020

Delhi Public School, Pune

10th CBSE, AISSE

CGPA: 9.20

2015 - 2017

(Holding a valid Indian Passport)